MIL-S-16104B
27 December 1983
SUPERSEDING
MIL-I-16104A
12 September 1951
(See 6.7)

MILITARY SPECIFICATION

SYNCHROSCOPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers switchboard-mounted synchroscopes which indicate phase angle between two alternating current (a.c.) power sources or show within limits their relative departure from synchronism.

1.2 Classification. The type designation of synchroscopes shall be in the following form, as specified (see 6.2.1).

MR	4R	W	060	SY	H
Component	Style	Color scheme	Frequency response	Unit	Special features
1.2.1	1.2.2	1.2.3	1.2.4	1.2.5	1.2.6

1.2.1 Component. Synchroscopes covered by this specification are designated by the two-letter symbol "MR".

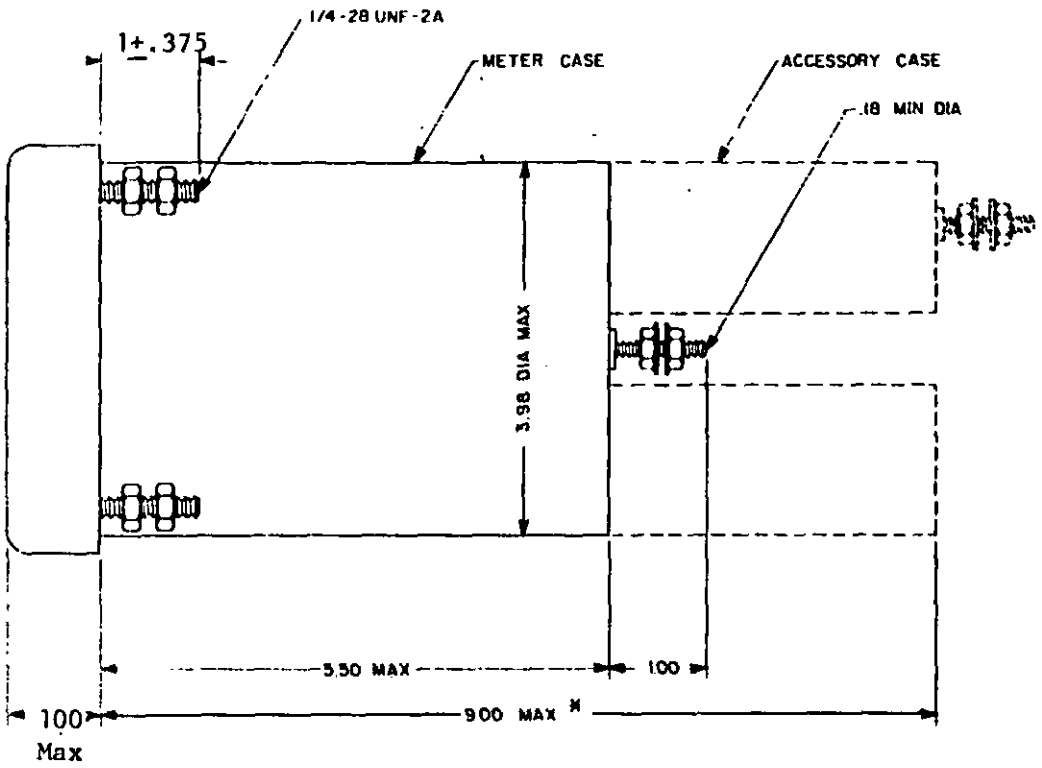
1.2.2 Style. The style is designated by a two-digit number which signifies the size and shape as follows: (See figures 1 and 2. Rounded-flanged synchroscopes shall be in accordance with the applicable plates of ANSI C39.1.)

1.2.2.1 First digit.

4: 4-1/2 inch synchroscope
6: 6 inch synchroscope

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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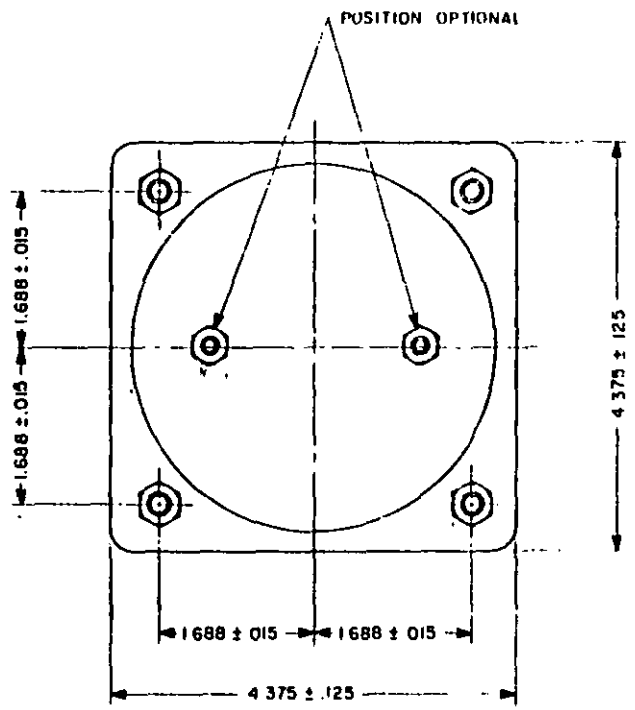


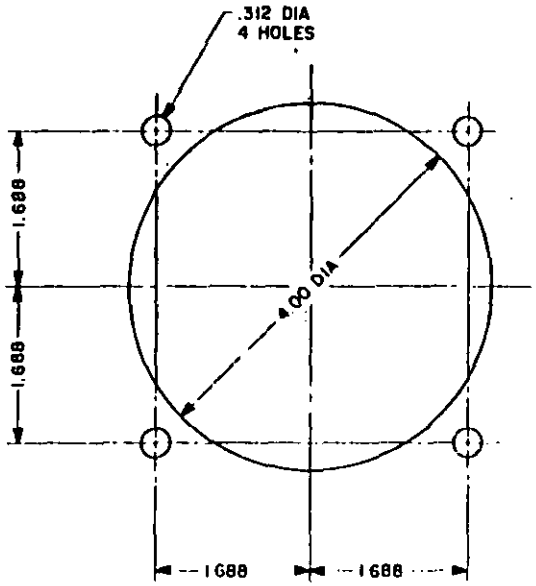
ALL DIMENSIONS ARE IN INCHES

* WITH INTERNAL ACCESSORIES

SH 12083

FIGURE 1. Rectangular meter (4-1/2 inches).





PANEL DRILLING PLAN

MIL-S-16104B

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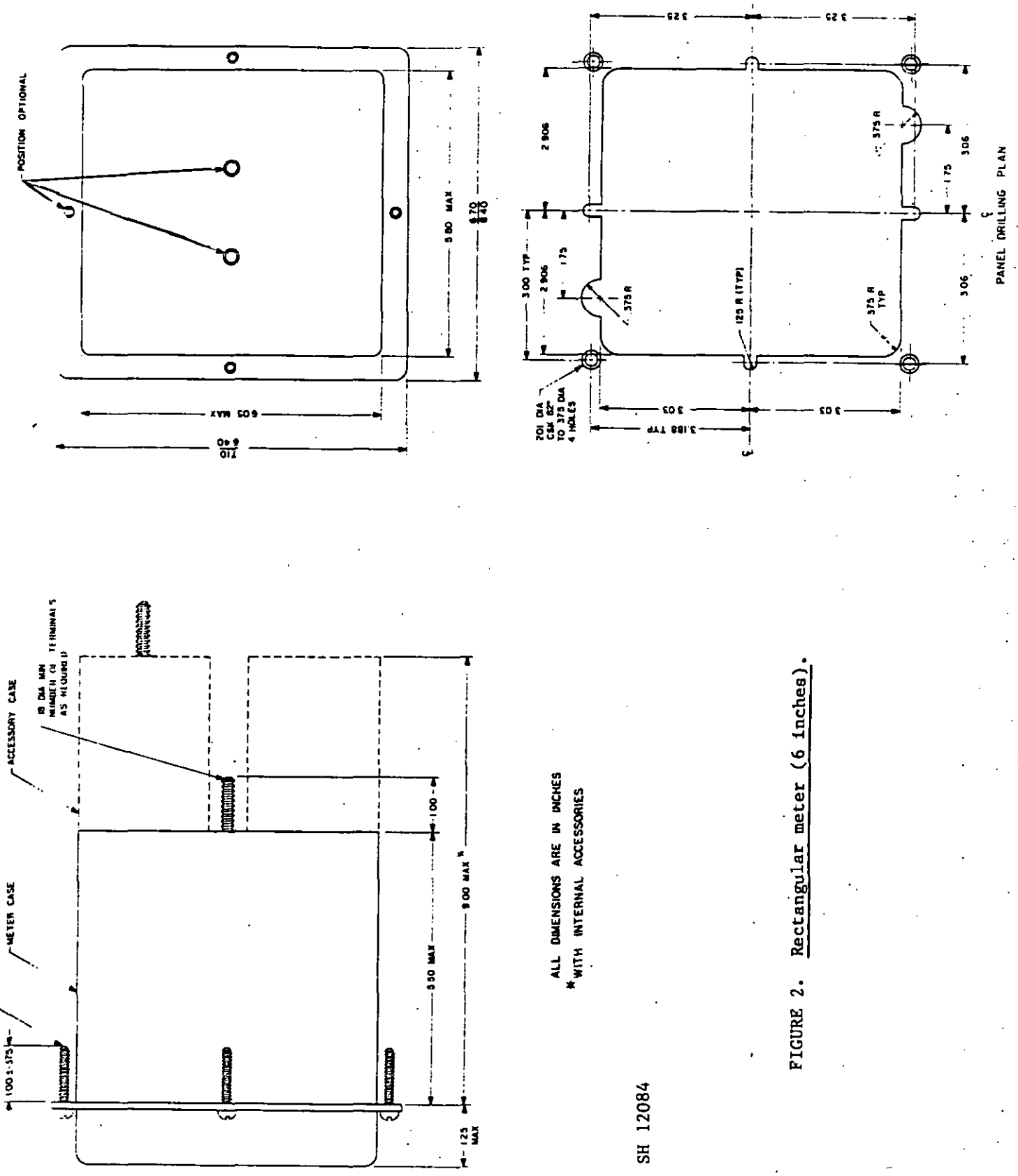


FIGURE 2. Rectangular meter (6 inches).

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1.2.2.2 Second digit.

R: Round flange
S: Rectangular flange

1.2.3 Color scheme.

B: Black dial, white pointer and lettering
W: White dial, black pointer and lettering

1.2.4 Frequency response. The frequency response shall be designated by a three digit number in hertz (Hz), zero-filled to the left.

1.2.5 Unit. The unit shall be "SY" for synchroscope.

1.2.6 Special features. (When applicable.)

H: High-impact (H.I.) shock

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-387	- Plastic Sheet, Laminated, Thermosetting (For Designation Plates).

MILITARY

MIL-M-14	- Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-W-80	- Window, Observation, Acrylic Base, Antielectrostatic, Transparent (for Indicating Instrument).
MIL-P-116	- Preservation, Methods of.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-G-3787	- Glass, Laminated, Flat (Except Aircraft).
MIL-E-15090	- Enamel, Equipment, Light-Gray, Formula No. 111.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-167-1 - Mechanical Vibration of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
C39.1 - Requirements for Electrical Analog Indicating Instruments.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. Synchroscopes furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Safety. The safety requirements shall be in accordance with the requirements of MIL-E-917.

3.3 Material. Synchroscopes shall be made of the material specified herein. However, when no definite material is specified, a material which will enable the synchroscopes to meet the performance requirements of this specification shall be used. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

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3.3.1 Restricted materials. Material restricted for Navy use by MIL-E-917 shall not be used in the construction of synchroscopes.

3.3.2 Recovered materials. Unless otherwise specified herein, all equipment and material incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3.3 Plastics. When applicable, the following types of plastics shall be used for construction of synchroscopes.

(a) Molded thermosetting plastic shall be in accordance with MIL-M-14.
(b) Laminated thermosetting plastic shall be in accordance with L-P-387.
(c) Plastic windows shall be in accordance with MIL-W-80.

3.3.4 Metals. Metals and the treatment of metals shall be corrosion-resistant in accordance with the requirements of MIL-E-917.

3.3.5 Glass. Glass shall be of the shatterproof type in accordance with class 1, type I of MIL-G-3787.

3.4 Operating voltage and frequency. The synchroscope shall operate with a.c. voltage between 105 and 125 volts (V) at the specified frequency under the reference conditions as specified in 4.3.1.

3.5 Sensitivity. The synchroscopes shall be sensitive to a difference of frequency between two a.c. sources up to 1.5 Hz and be within 1 percent accuracy up to 0.1 Hz.

3.6 Cases. Synchroscope cases shall be constructed only from the materials specified herein. The cases shall be constructed with close-fitting joints to minimize the entrance of dust and moisture. Metal cases shall be protected against corrosion in accordance with MIL-E-917.

3.7 Windows. Synchroscopes shall be provided with a glass or plastic window. Regardless of the type, it shall be free from discoloration, scratches, striae, or other defects which may interfere in any way with the normal reading of the indicator in daylight. When required by the applicable specification sheet, spraytight and watertight windows shall conform to MIL-STD-108.

3.8 Dial and pointer.

3.8.1 Dial. The dial shall be readily removable and made of stiff material firmly secured to the mounting. The material and construction shall insure freedom from warping, fading, or other deterioration during the tests specified in section 4. Paper dials will be acceptable only when adequately bonded to a metal support.

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3.8.1.1 Scale markings. Scale markings shall indicate when one source of a.c. is synchronous with the one to be synchronized and the speed of the incoming source to the one with which it is being synchronized. The words "slow" and "fast" shall indicate the speed of the incoming source with the dial pointer rotating in a counterclockwise direction for the former and a clockwise direction for the latter.

3.8.2 Pointer. The pointer shall be rigidly constructed and shall not become damaged or distorted when subjected to any of the tests of this specification.

3.9 Finish. The portion of the case exposed to view from the front of the panel shall have a light gray finish in accordance with formula no. 111, class 2 of MIL-E-15090. Light reflecting trimmings shall not be used. Metal cases shall be rendered suitably resistant to corrosion before the application of the final gray finish.

3.10 Identification plates. Identification and other marking plates shall be in accordance with MIL-STD-130.

3.11 Dielectric strength. Synchroscopes shall withstand the dielectric test specified in 4.7.2 without damage or flashover. A test voltage of 1900 V root mean square (Vrms) shall be applied as follows:

(a) External live parts to external shield of the case.
(b) External live parts to exposed metal parts on the front of indicator.
(c) External live parts to metal mounting panel.

3.12 Insulation resistance. When tested as specified in 4.7.3, the insulation resistance of synchroscopes shall be at least 20 megohms.

3.13 Vibration. Synchroscopes shall withstand the vibration test specified in 4.7.4 without permanent change in indication greater than 1 percent of total deflection.

3.14 Moisture resistance. Synchroscopes shall operate freely following exposure to the moisture resistance test specified in 4.7.5.

3.15 Mounting. Mounting shall be in accordance with figures 1 or 2, or ANSI C39.1 as applicable. Attaching hardware shall be supplied.

3.16 Shock. When required (see 6.2.1), the synchroscopes shall be constructed to withstand the shock test specified in 4.7.6 or 4.7.6.1.

3.17 Voltage influence. A 7 percent voltage variation on both elements shall not produce a change of indication. A 1 percent voltage difference shall not change the indication by more than 2 degrees (see 4.7.7).

3.18 Overvoltage. The synchroscopes shall sustain the overvoltage test as specified in 4.7.8.

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3.19 Life cycle. The synchroscopes shall be made to withstand the life cycle test as specified in 4.7.9. The synchroscope shall not overheat or change accuracy beyond 1 percent (3-1/2 degrees).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system and a calibration system in accordance with the data ordering document (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.4).
(b) Comparison inspection (see 4.5).
(c) Quality conformance inspection (see 4.6).

4.3 Inspection conditions.

4.3.1 Reference conditions. Unless otherwise specified, all inspections shall be performed in accordance with the reference conditions specified in 3.4 and as follows:

(a) Temperature: 23 ± 2 degrees Celsius (°C).
(b) Relative humidity: Greater than 50 percent.

4.3.2 Thermal stability. Unless specified herein, synchroscopes shall be maintained at the specified inspection temperature for not less than 2 hours to attain thermal stability before taking readings.

4.4 Qualification inspection. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). Qualification examination and inspection shall consist of the examination and tests specified in table I in the order shown.

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4.4.1 Sample for qualification inspection. Three synchroscopes of each type designation shall be submitted for the examination and tests specified in table I.

TABLE I. Qualification, comparison and quality conformance inspection.

Inspection	Qualification inspection	Comparison inspection	Quality conformance inspection	Requirement paragraph	Inspection paragraph
Visual inspection	X	X	X	3.3, 3.6 thru 3.10 3.15 and 3.22	4.6.3
Sensitivity	X	X	X	3.5	4.7.1
Operating voltage and frequency	X	X	X	3.4	4.7.1
Dielectric strength	X	X		3.11	4.7.2
Insulation resistance	X	X		3.12	4.7.3
Vibration	X	-		3.13	4.7.4
Moisture resistance	X	-		3.14	4.7.5
Shock	X	X		3.16	4.7.6
Voltage influence	X	X		3.17	4.7.7
Overvoltage	X	X		3.18	4.7.8
Life cycle	X	-		3.19	4.7.9

4.5 Comparison inspection. Within 4 years after qualification inspection and each successive 4-year period in which the synchroscope is offered for delivery thereafter, one synchroscope shall be subjected to the comparison inspection specified in table I.

4.6 Quality conformance inspection. Each production synchroscope shall be subjected to the quality conformance inspection specified in table I.

4.6.1 Inspection lot. All synchroscopes of the same type presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.6.2 Inspection sample. The inspection sample shall be the single sampling plan for normal inspection in accordance with MIL-STD-105. The acceptance quality level (AQL) shall be 1.0 percent defective.

4.6.3 Visual inspection. Each of the synchroscopes shall be examined by the inspector to verify compliance with this specification. Any synchroscope containing one or more visual defects shall be rejected.

4.7 Methods of inspection.

4.7.1 Sensitivity. Each synchroscope shall be tested for compliance with 3.5 by connecting it to two a.c. sources (see 3.4), one of which shall be slowly increased until the synchroscope ceases to operate and then slowly decreased in the opposite direction until the synchroscope again ceases to operate and then slowly returned to 0 frequency difference. This test shall be repeated with the inputs interchanged.

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4.7.2 Dielectric strength. Synchroscopes shall be mounted on steel panels by the mounting screws furnished with the indicators. The synchroscopes shall be subjected to a voltage of the rms value specified in 3.11 applied between all external live parts of the indicator and the following:

(a) The external shield of the case.
(b) Exposed metal parts on the front of the synchroscope.
(c) The metal panel on which the synchroscope is mounted.

The frequency shall be nominal 60 Hz and shall approximate as closely as possible a true sine wave in form. The voltage shall be applied continuously for 60 seconds. The test voltage shall be measured by the voltmeter method whereby the measuring meter derives its voltage directly from the circuit or indirectly via instrument transformer. To avoid voltage surges, the applied voltage shall be raised to its full value gradually and, upon completion of the 60-second test, gradually reduced.

4.7.3 Insulation resistance. The synchroscope shall be mounted on a steel panel. Using 500 V direct current, the insulation resistance (see 3.12) shall be determined between all external live parts and the following:

(a) The external shield of the case.
(b) Exposed parts on the front of the synchroscope.
(c) The metal panel on which the synchroscope is mounted.

4.7.4 Vibration. Synchroscopes shall be subjected to the type I tests of MIL-STD-167-1. The changes in indication shall then be determined (see 3.13).

4.7.5 Moisture resistance. The synchroscope shall be placed in a humidity chamber and kept for 6 hours at a temperature of 65°C and a relative humidity of 90 percent. Electrical energy shall be supplied to the synchroscope. At the end of the 6-hour period, the heat shall be shut off and the meter deenergized. The synchroscope shall then be permitted to cool for 18 hours in this atmosphere, the relative humidity of which rises as the temperature declines. (Precaution shall be taken to prevent dripping on the synchroscope.) A temperature of 35°C or lower, but not less than 25°C, shall be attained during the cooling period. The complete cycle shall then be repeated. Upon completion of the second cycle, the synchroscope shall be removed from the humidity chamber and permitted to dry for 24 hours at room temperature with the relative humidity less than 60 percent. Synchroscope readings shall then be taken at the reference temperature. The meter window shall be removed, and the synchroscope shall then be subjected to two additional cycles. The synchroscope shall be removed from the humidity chamber and permitted to dry for 24 hours at room temperature with the relative humidity less than 60 percent. Meter readings shall then be taken at reference temperature. Synchroscope readings shall be within specified tolerances. Only one synchroscope of the sample shall be subjected to this test (see 3.14).

4.7.6 Shock. Two synchroscopes shall be tested for shock in accordance with ANSI C39.1.

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4.7.6.1 Shock (H.I. shock type only). One synchroscope shall be tested in accordance with the grade A, class 1, type B shock test specified in MIL-S-901. The features of the test shall be as follows:

(a) Weight designation of the shock test: Lightweight.
(b) Definition of failure to perform principal functions:

(1) Loosening, cracking, or damaging of any screws, bearings, pivots, transparent cover or other parts.
(2) Appreciable distortion of any parts.
(3) Inability to indicate synchronism.
(4) Impairment of specified sensitivity.

(c) Acceptable method of mounting the equipment on the shock-testing machine: In accordance with MIL-S-901.
(d) Place at which shock test will be conducted: At a Government laboratory specified by the command or agency concerned.
(e) Disposal of shock-tested equipment: Equipment which has been subjected to the class H.I. shock test and has passed this test shall be considered acceptable for such service as the command or agency concerned may authorize.

4.7.7 Voltage influence. Each synchroscope shall be tested for voltage influence in accordance with 3.17 and meet the requirements specified therein.

4.7.8 Overvoltage. Each synchroscope shall be energized at nominal voltage. The frequencies shall be mismatched so as to cause a rotation of one revolution per 10 to 20 seconds. The voltage shall be set on both elements at 125 V and then one element raised at 170 V and held for 2 minutes. The synchroscope shall be deenergized and the test repeated, energizing the other element to 170 V. The meter shall then be subjected to the quality conformance tests (see 3.18).

4.7.9 Life cycle. Each synchroscope shall be cycled for 720 hours, approximately 4 minutes on and 1 minute off. The speed of rotation shall be a minimum of one revolution per 10 seconds. The indicator shall be monitored during one cycle of each 24-hour period of operation and perform as outlined in 3.19.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

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5.1 Preservation and packaging. Synchroscopes and mounting hardware shall be individually cushioned and unit-packaged in accordance with method IIb of MIL-P-116, without the use of contact preservatives. The packaging shall conform to level A or C of MIL-E-17555, as specified (see 6.2.1).

5.2 Packing. Packing shall be in accordance with level A or C of MIL-E-17555, as specified (see 6.2.1).

5.2.1 For domestic shipment-immediate use. The packing for domestic shipment-immediate use synchroscopes shall conform to level C of MIL-E-17555.

5.2.2 For domestic shipment and storage or overseas shipment. Synchroscopes packaged for domestic shipment and storage or for overseas shipment shall conform to level A of MIL-E-17555.

5.3 Marking. In addition to any special marking required (see 6.2.1), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Synchroscopes covered by this specification are intended for switchboard use in general application of electrical measurement, including shipboard and land installation. Type MR4SW060SYH (shockproof) synchroscopes are normally used in combatant Naval ships.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) When synchroscopes shall withstand shock (see 3.16).
(d) Level of packing, preservation and marking (see 5.1, 5.2 and 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1.1	Inspection system program plan	DI-R-4803	----
4.1.1	Calibration system description	DI-R-7064	10.2

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-16104 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Definition. For the purpose of this specification, the definitions of American National Standard C39.1 and the following definition should apply:

6.6.1 Synchronism. Synchronism is the condition existing between two or more periodic quantities of the same period when the phase difference between them is zero.

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army -CR
Navy - SH

Review activity:
Navy - MC

User activities:
Army - ME
Navy - AS

Preparing activity:
Navy - SH
(Project 6625-0603)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-16104B	SYNCHROSCOPE

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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